
A retrieval augmented generation model with reasoning for the WCPFC tuna conservation management measures

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What is a RAG model?



RAG = RETRIEVAL
AUGMENTED GENERATION



RAG MODELS COMBINE
LARGE LANGUAGE MODELS
WITH CHAT INTERFACES
WITH A PROPRIETARY
DATABASE

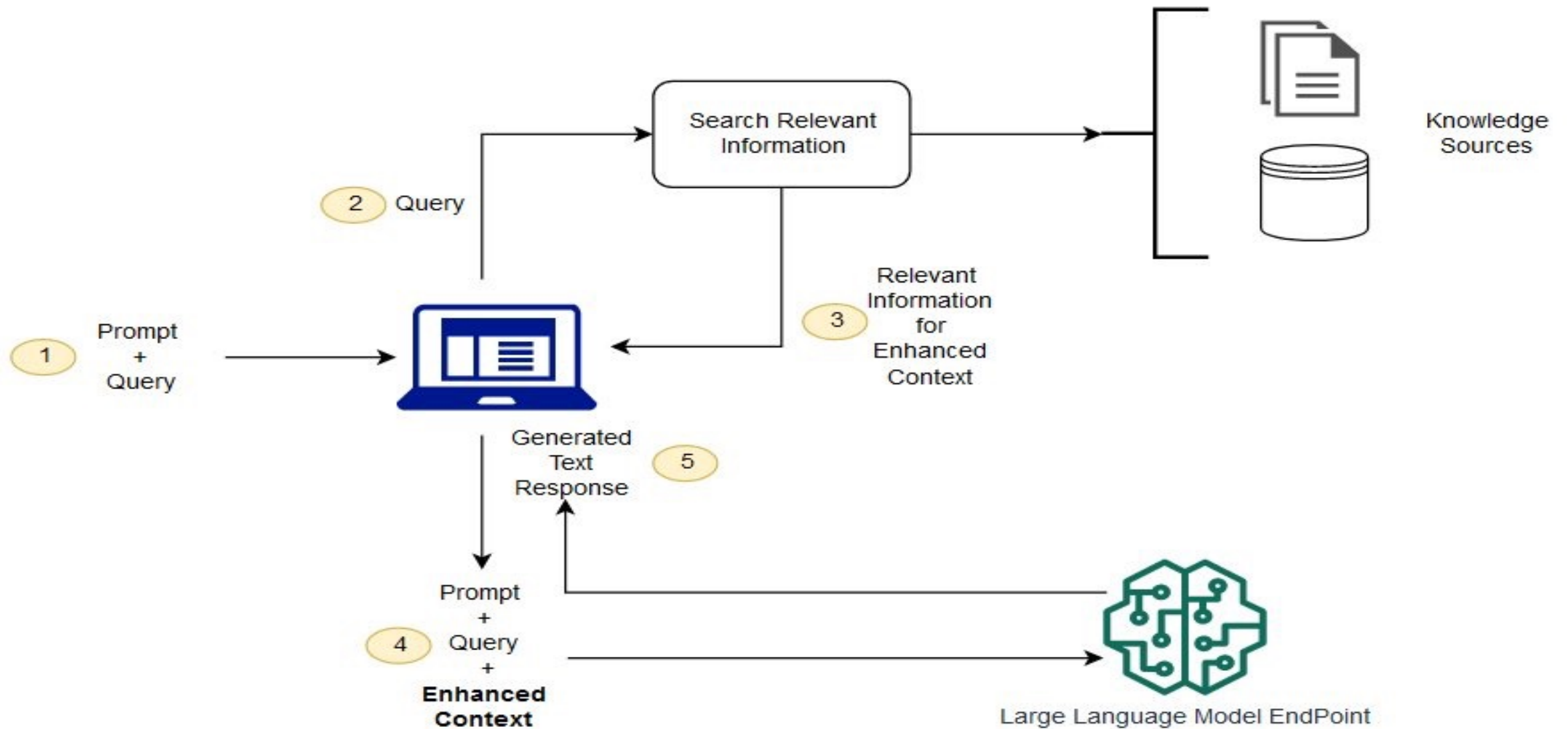


THIS MEANS YOU CAN
SECURE THE DATA RATHER
THAN SHARING IT WITH
THIRD PARTIES



RAG MODELS ARE ONE WAY
IN WHICH AI (ARTIFICIAL
INTELLIGENCE) IS
SPREADING





Source: <https://docs.aws.amazon.com/sagemaker/latest/dg/jumpstart-foundation-models-customize-rag.html>

Introducing the AI Fisheries Manager

AI Fisheries Manager 🐟



Ask the fisheries manager a question

Data

- We used public data (texts) from the Region or relevant to the Region that would form part of the knowledge base for experienced fisheries managers
- The focus is on Pacific Tuna Fisheries
- Examples include: UNCLOS, the UN Fish stocks agreement, the WCPFC conservation management measures, various reports including stock assessment reports

Data storage and use

- The text documents are converted to numerical form (as vectors) and stored in a vector store (think of this as a database)
- We store this in memory on the computer that runs the application so no data leaves the premises
- The user that is someone in this audience hopefully
- Writes a question in the search box
- The data base is then searched using semantic search (it matches against similarity)
- It returns an answer that will hopefully support decision-making either for individuals or for groups (e.g. meetings)
- One can think of it as a combination of a Knowledge Base and a Decision Support Tool

Example 1



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Ask the fisheries manager a question

Summarize the relevant Bigeye CMMs

Reply: **Interim Bigeye Tuna Catch Limits (paragraphs 38-40):** CCMs listed in Table 3 of Attachment 1 must limit their bigeye tuna catches to the specified levels. Any overages will be deducted from the following year's catch limit. CCMs must report monthly bigeye catches to the Secretariat, which will notify all CCMs when 90% of a CCM's catch limit is reached. These limits do *not* represent allocation of fishing rights.

Bigeye Tuna Catch Limit for Members with Low Historical Catches (paragraph 41): Members who caught less than 2,000 tonnes of bigeye in 2004 must ensure their annual catch doesn't exceed 2,000 tonnes.

Transition to Equitable Allocation Framework (paragraph 42): The Commission commits to transitioning to a more equitable allocation framework for bigeye fishing opportunities, considering Articles 8, 10(3), and 30 of the Convention. This process will begin in 2026, aiming for agreement on hard limits for all Members and Participating Territories.

Limits on Longline Vessels Targeting Bigeye Tuna (paragraphs 45-46): CCMs (excluding SIDS and Indonesia) cannot increase the number of their longline vessels (with or without freezing capacity) targeting bigeye tuna above the levels established under CMM 2013-01. This excludes CCMs with domestic quota systems, including individual transferable quotas, under a regulated management framework.

Example 2



Ask the fisheries manager a question

in which CMMs is the precautionary principle applied?

Reply: The precautionary principle is applied to all harvest strategies established pursuant to Conservation and Management Measure (CMM) 2022-03. This CMM, titled "CONSERVATION AND MANAGEMENT MEASURE ON ESTABLISHING A HARVEST STRATEGY FOR KEY FISHERIES AND STOCKS IN THE WESTERN AND CENTRAL PACIFIC OCEAN," explicitly references the precautionary principle as outlined in the UN Fish Stocks Agreement, the Convention on the Conservation and Management of Highly Migratory Fish Stocks, and the FAO Code of Conduct for Responsible Fisheries. While the question asks which *CMMs*, the provided text identifies one CMM (2022-03) that applies the precautionary approach as a core principle for developing and implementing harvest strategies for *all* key fisheries and stocks under the purview of the WCPFC.

Example 3

AI Fisheries Manager 🐟



Ask the fisheries manager a question

Tell me about harvest strategies

Reply: Harvest strategies are a best-practice approach to fisheries management decision-making. They are proactive and adaptive, providing a framework for using the best available information about a stock or fishery and applying an evidence- and risk-based approach to setting harvest levels. This creates a more stable operating environment where management decisions are more consistent, predictable, and transparent.

Key elements of a harvest strategy include:

- **Management Objectives:** Defined operational objectives, including timeframes, for the fishery or stock. These consider biological, ecological, economic, and/or social goals, balancing any competing objectives.
- **Reference Points:** Target and limit reference points for each stock. Targets aim to meet management objectives, while limits constrain harvesting within safe biological boundaries.
- **Acceptable Levels of Risk:** Defined acceptable levels of risk, particularly the very low risk of breaching

Conclusion

I can demo a live version on the side of the conference for those interested

Thanks for listening

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